Name : Lingesh M

Reg No : 411625104034

Department : B.E CSE

Subject : PYTHON

Batch : 02

Number Guessing Game using Python

Aim

The aim of this program is to design and implement a **Number Guessing Game** using Python.

The program generates a random number between 1 and 100, and the user must guess the correct number. The program provides hints whether the guess is too high or too low until the correct number is guessed.

Procedure

Step 1: Importing Required Module The

program begins by importing:

- random module to generate random numbers

Step 2: Displaying Welcome Message

The program prints a welcome message and game instructions.

Step 3: Generating Random Number

A random number between 1 and 100 is generated using:

number = random.randint(1, 100)

Step 4: Initializing Attempts Counter

A variable attempts is initialized to 0 to count the number of guesses made by the user.

Step 5: Using While Loop

A while True loop is used to continuously ask the user for input until the correct guess is made.

Step 6: Comparing the Guess

- If guess > number → Display "Too High"
- If guess < number → Display "Too Low"
- If guess == number → Display success message and break the loop

---

Step 7: Displaying Performance Rating

After guessing correctly, the program prints stars (*) equal to the number of attempts to show performance rating.

Code

```
 import random

print("  Welcome to Number Guessing Game  ")
print("-------------------------------------")
number = random.randint(1, 100)
attempts = 0 while True:
   guess = int(input("Enter your guess (1-100): "))
attempts += 1
if guess > number:
      print("Too High!   Try again.\n")
elif guess < number:
      print("Too Low!   Try again.\n")
 else:
      print("\n  Correct! You guessed the number.")
      print("Total Attempts:", attempts)
break

print("\nYour Performance Rating:")
for i in range(attempts):

  print("*", end=" ")
```

Program Output (Sample)

```
 Welcome to Number Guessing Game
-------------------------------------
Enter your guess (1-100): 69
Too High! Try again.

Enter your guess (1-100): 65
Too High! Try again.

Enter your guess (1-100): 50
Too High! Try again.

Enter your guess (1-100): 40
Too High! Try again.

Enter your guess (1-100): 20
Too High! Try again.

Enter your guess (1-100): 10
Too Low! Try again.

Enter your guess (1-100): 15
Too High! Try again.

Enter your guess (1-100): 12
Too High! Try again.

Enter your guess (1-100): 11

 Correct! You guessed the number.
Total Attempts: 9


...Program finished with exit code 0
Press ENTER to exit console.
```

Result

The Number Guessing Game was successfully implemented using Python.

The program correctly generates a random number, accepts user guesses, provides hints, and calculates the number of attempts.

It effectively demonstrates:

- Random number generation
- While loop
- Conditional statements

- User input handling